



Physics-Informed Machine Learning for Cross-Surface Generalization in Droplet Impact Prediction

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1 The question

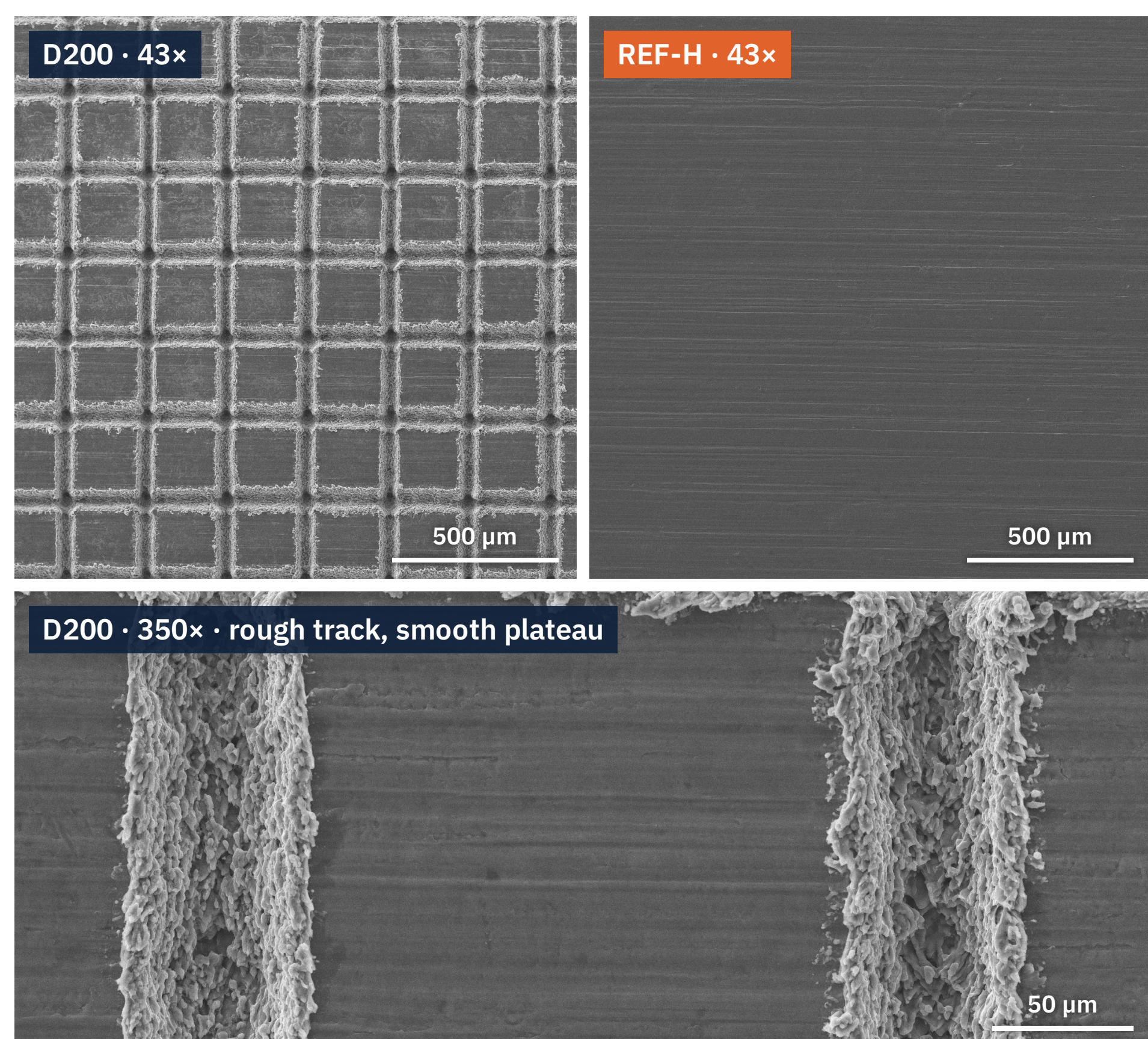
Can a model trained on 12 laser-textured surfaces predict how far a droplet spreads on a surface it has never seen?

Droplet-impact models are usually scored on random splits, where every surface is in training. Surface design needs the opposite: a prediction **before** the surface is made.

1,498 impacts on 12 textured aluminium surfaces	125 impacts on the smooth plate REF-H	1–160 mPa·s, water to 91 wt% glycerol
8–119 Weber number	9–4,435 Reynolds number	50–800 µm channel pitch; depth 6 or 25 µm

Public dataset of Može et al. [1], University of Ljubljana: nanosecond-laser channels on aluminium 1050A, 5,000 fps imaging. Target: maximum spreading $\beta_{\max} = D_{\max}/D_0$.

2 Reading the surface from SEM

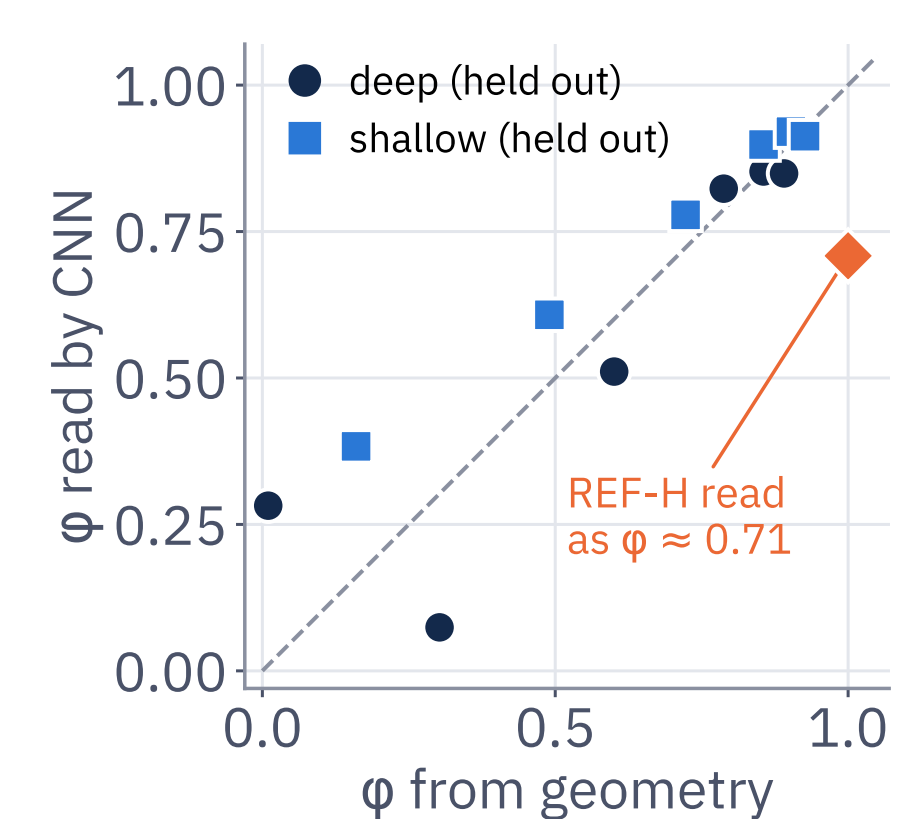


Plateaus between laser tracks look like the smooth plate. Track widths from SEM ($\approx 45 \mu\text{m}$ deep, $30 \mu\text{m}$ shallow) give every surface two descriptors:

$$\varphi = (\max(s - w, 0) / s)^2, \quad V_{\text{tex}} = (1 - \varphi) h$$

Textured φ runs from 0.01 (D50) to 0.93 (S800); **REF-H is simply $\varphi = 1$** . Invented spacing/depth codes moved REF-H error by up to 0.022; with φ , $\pm 10 \mu\text{m}$ in track width moves it by at most 0.005.

3 From SEM image to prediction



A 9,070-parameter CNN reads three SEM views (43x, 100x, 350x) of a surface **it never saw** and estimates φ and V_{tex} . The GP on these gives **0.038** vs 0.040 on unseen surfaces, better on 8 of 12, but not significant. REF-H is read as $\varphi \approx 0.71$: no training surface is smoother than 0.93.

Next steps

- Measured advancing and receding angles for every fluid–surface pair as an input.
- Noise that depends on viscosity (Ohnesorge number).
- New surfaces, with a test set held back before any tuning.

4 Physics inputs

$$Re = \frac{\rho V D_0}{\mu} \quad We = \frac{\rho V^2 D_0}{\sigma} \quad \beta_{\max} Re^{-1/5} = \frac{\sqrt{P}}{A + \sqrt{P}}$$

Recomputed Re and We match the dataset to 5×10^{-7} ; Oh adds nothing new. The Laan scaling law [3] ($P = We Re^{-2/5}$) was derived on wettable surfaces: here it reaches only **$R^2 = 0.71$** and misses unseen surfaces by **11%** on average.

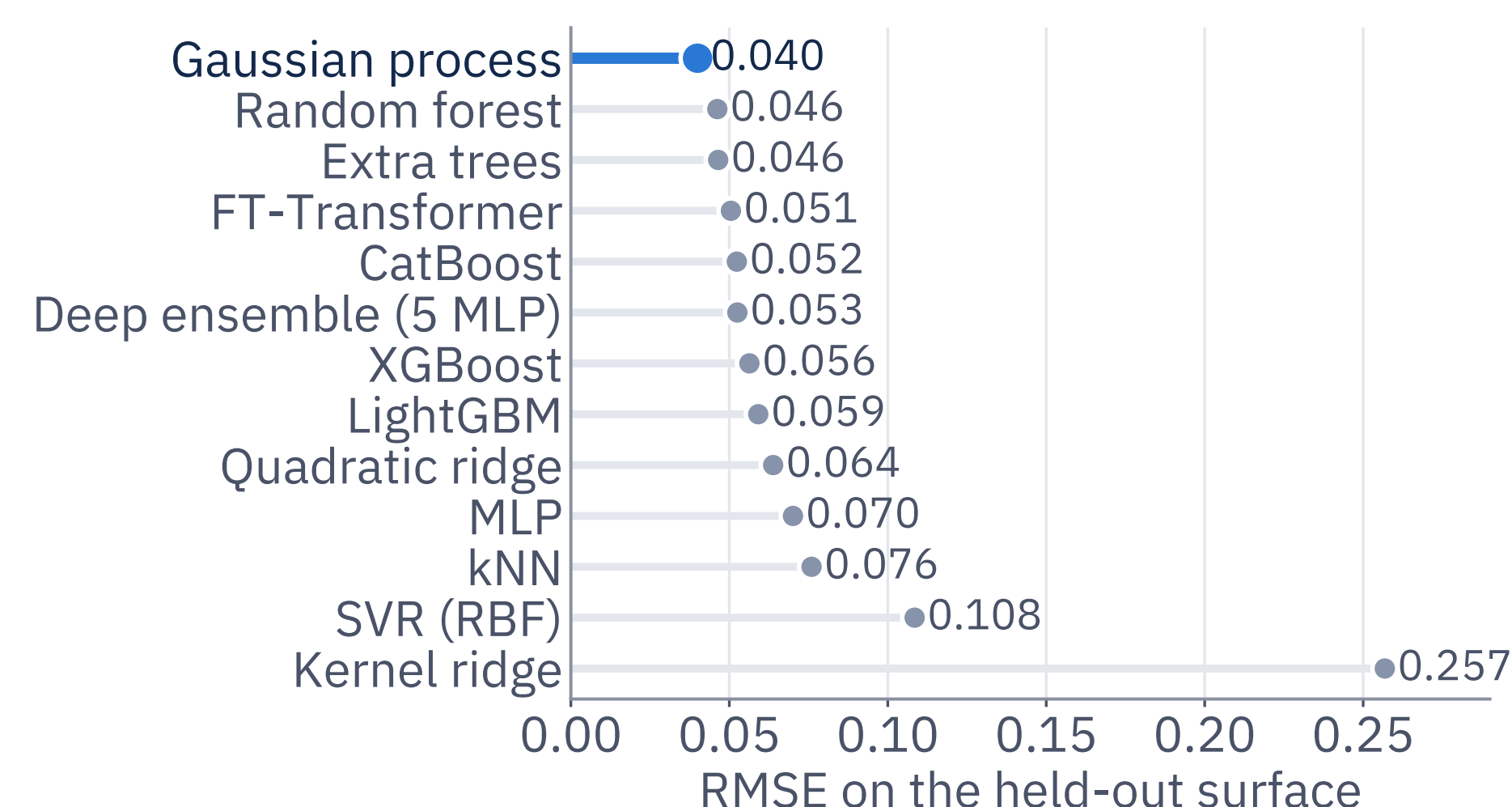
5 Leakage-safe evaluation

Each condition was repeated five times, so a random split puts near-identical twins in train and test.

- Grouped 5-fold CV** over 297 replicate groups: accuracy on seen surfaces.
- Leave-one-surface-out**, 12 folds: the model-selection criterion.
- REF-H held out**: never used to train or choose a model (stress test).

Inputs for every model: $\ln Re$, $\ln We$, $\ln D_0$, φ , V_{tex} ; target $\ln \beta_{\max}$. A change is accepted only if its 95% interval, from **resampling whole surfaces**, excludes zero.

6 13 models, one test



0.036

GP error, seen surfaces ($R^2 = 0.993$)

0.040

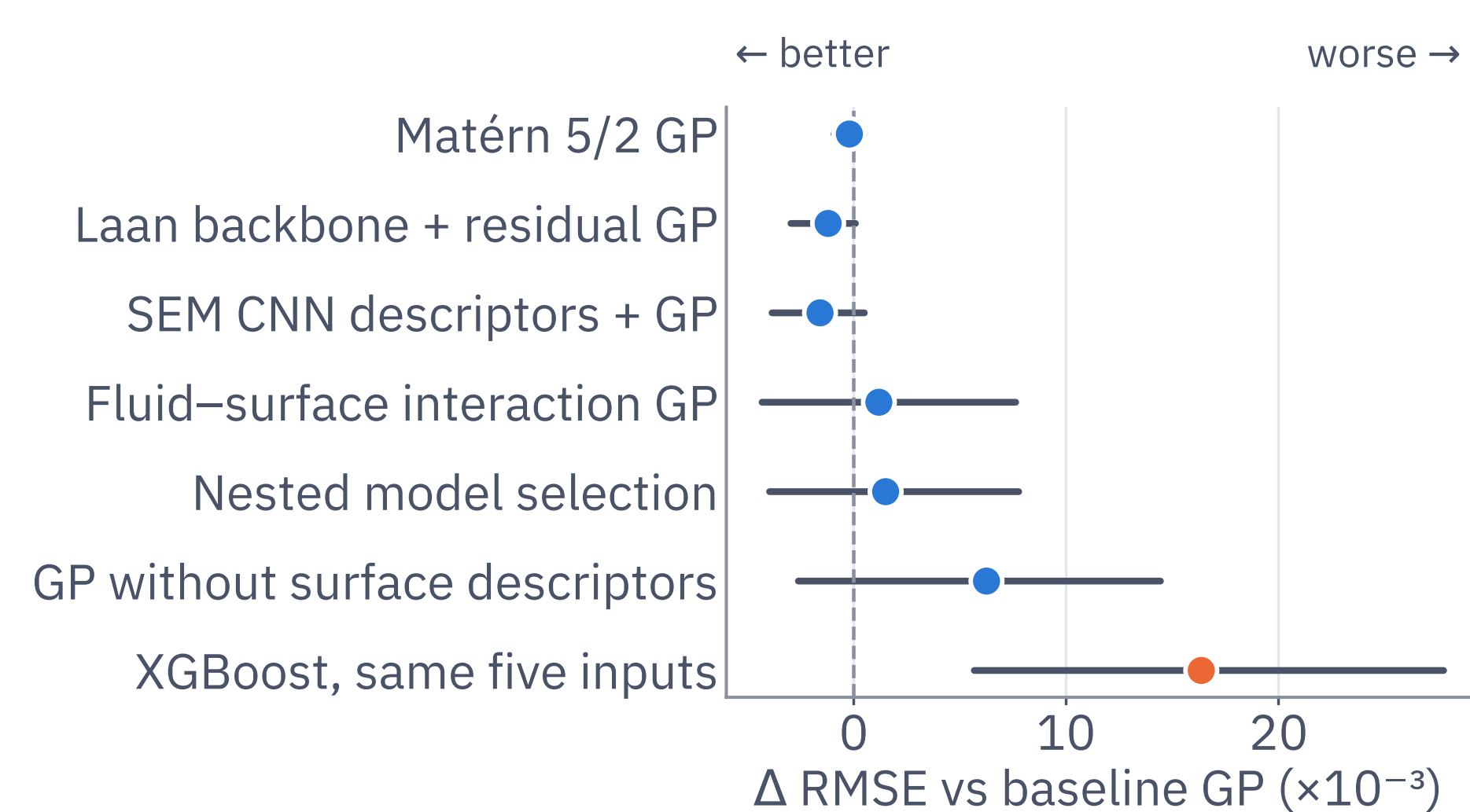
GP error, unseen surface (1.1% mean abs. error)

0.056

XGBoost, same inputs, unseen surface

Gaussian process (Matérn 3/2, one length scale per input): 29% lower error than XGBoost on unseen surfaces (95% CI of difference 0.006–0.028). Every learner's hardest surface is D50, the densest texture.

7 What did not help



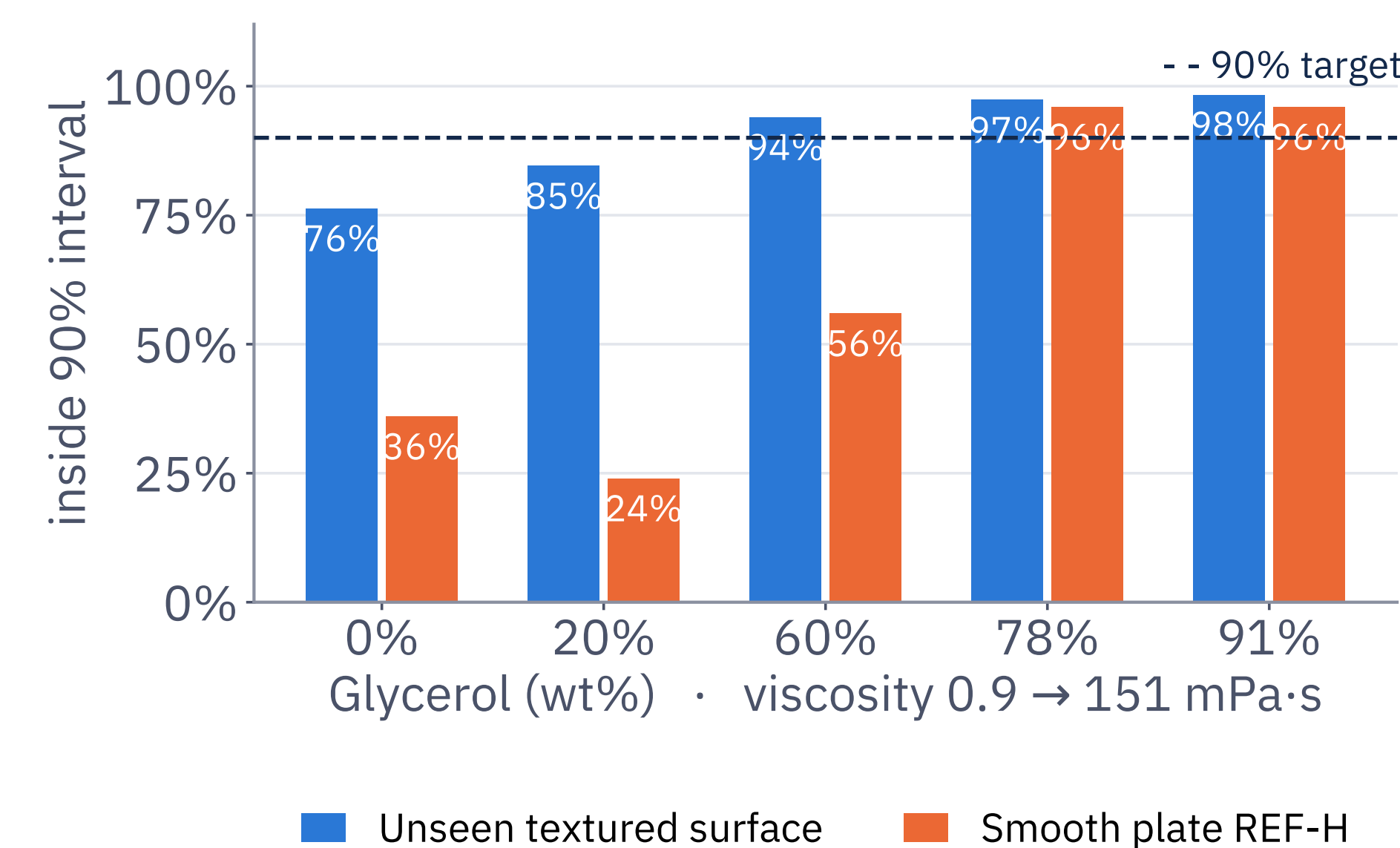
Paired surface-bootstrap 95% intervals (4,000 draws). Only XGBoost is significantly different (worse). The Laan law alone is off the scale ($+210 \times 10^{-3}$).

A **Laan-law backbone**, other kernels, nested selection and SEM-CNN descriptors each change the error by under 0.002. Removing φ and V_{tex} : $0.040 \rightarrow 0.046$ (REF-H $0.059 \rightarrow 0.069$), mostly on D50; with 12 surfaces, a hint, not proof. **The simple GP stays.**

Try it

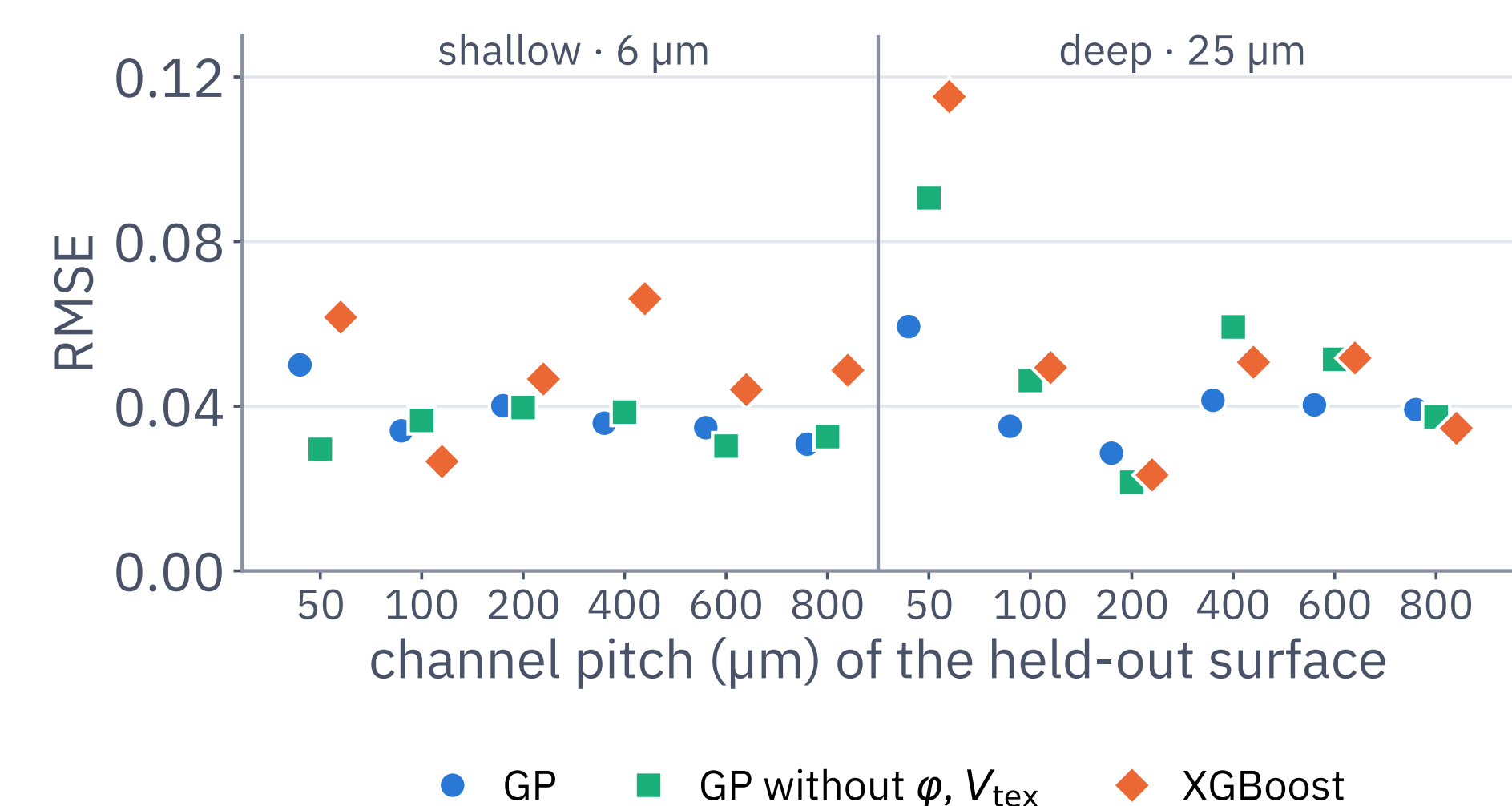
Python API and browser app with all five trained GPs. Every prediction carries a 90% interval and flags extrapolation and smooth surfaces. Browser and Python agree to 3×10^{-12} ; 18 tests pass.
`python -m droplet.serve`

8 Knowing when not to trust it



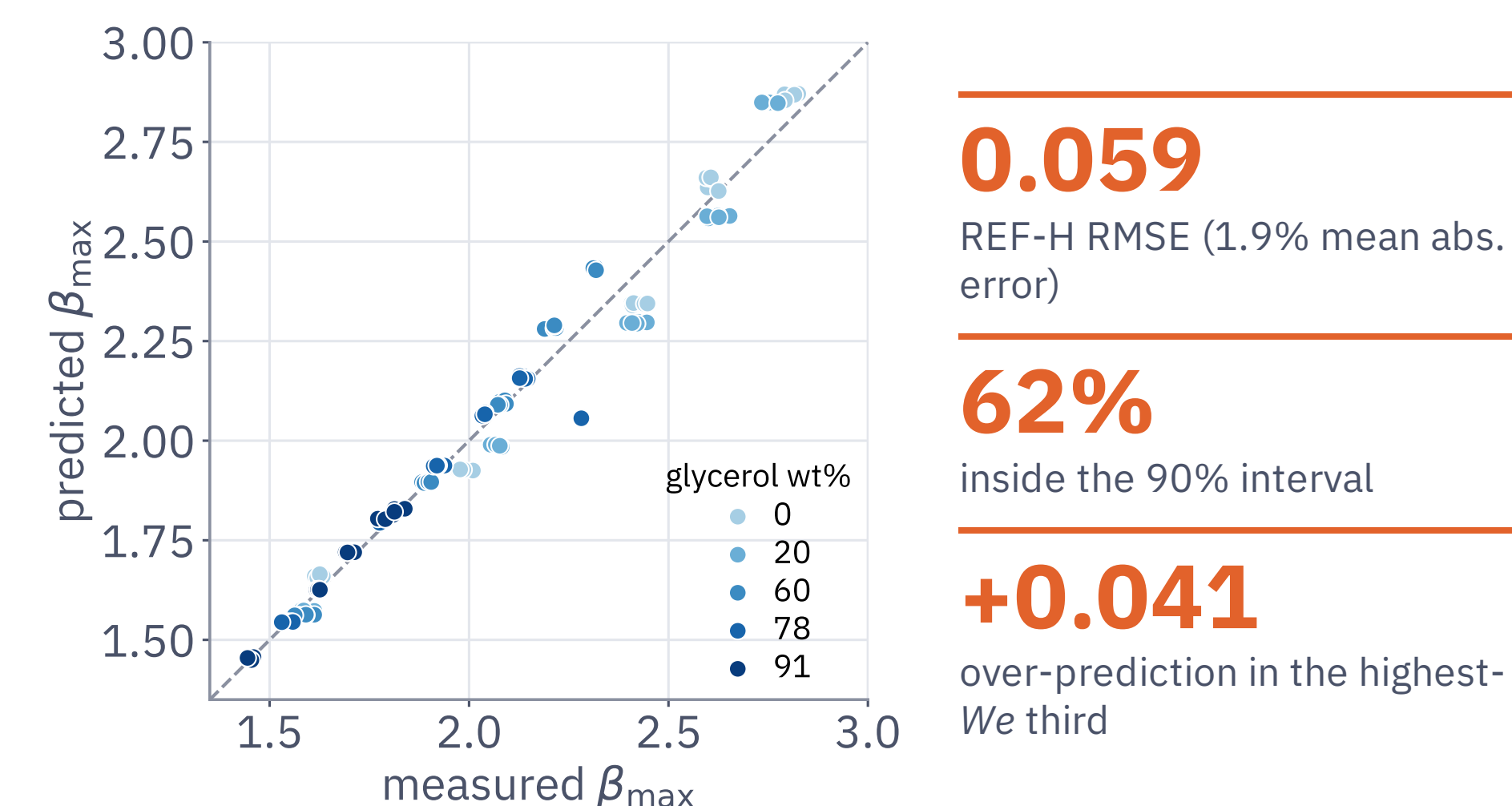
On unseen surfaces, **90.1%** of impacts fall inside the 90% interval (84–96% per surface; 92.6% with nested calibration). But one noise level does not fit all fluids: **76%** for water, **98%** for 91 wt% glycerol. Thin liquids feel the surface more. The OOD check flags D50 and S50, the hardest surfaces.

9 Surface by surface



D50 ($\varphi = 0.01$, the edge of the range) is hardest for every learner. The descriptors cut its error from 0.091 to 0.059, but hurt on S50 and D200.

10 Smooth-plate stress test



0.059

REF-H RMSE (1.9% mean abs. error)

62%

inside the 90% interval

+0.041

over-prediction in the highest-We third

Misses concentrate at high We and in water, 20% and 60% glycerol (24–56% coverage); viscous fluids stay at 96%. REF-H is hydrophobic; the textures are superhydrophobic (water advancing angle $162\text{--}167^\circ$). **Wettability is the missing input.** Less friction over trapped air may explain why textures spread further **HYPOTHESIS**

★ Take-aways

- Judge droplet models on **unseen surfaces**. Random splits hide the gap.
- Describe surfaces with quantities every surface has: the plateau fraction φ makes the smooth plate the limit of the textured family.
- A **Gaussian process** beats 12 alternatives on unseen surfaces (0.040); physics backbones and bigger models add no significant gain.
- Intervals are calibrated on average, not per fluid, and fail on a different wettability. **Say so on the tool.**

References

- M. Može et al., Data in Brief 61 (2025) 111697; data doi:10.17632/wsh8rxwd38.1
- S. Jereb et al., Biomimetics 10 (2025) 357 (random-split GP on the same data)
- N. Laan et al., Phys. Rev. Applied 2 (2014) 044018